

DATE:

EX.NO: 2b

STUDY OF SHELL PROGRAMMING

AIM:

To study shell programming and implement several programs.

(i) Sum of two numbers:

Algorithm:

1. Start
2. Read two number from user
3. Add the given two numbers.
4. Print the result
5. Stop

Code:

```
echo "Enter the first number: " read a
echo "Enter the second number: " read b
c=$((a+b))
echo "Sum of $a and $b is $c"
```

Output:

```
[CS24172 OSlab]$ vi sum.sh
```

```
[CS24172 OSlab]$ sh sum.sh
```

Enter the first number:

20

Enter the second number: 30

Sum of 20 and 30 is 50

(ii) Swapping of two numbers:

Algorithm:

1. Start
2. Read two numbers from the user
3. Swap numbers using temporary variable
4. Print the result
5. Stop

Code:

```
echo "Enter first number: " read fr
echo "Enter second number: " read sec
echo "Before swap" echo
"first=$fr" echo
"second=$sec" temp=$fr
fr=$sec sec=$temp
echo "After swap" echo
"first=$fr" echo
"second=$sec"
```

Output:

```
[CS24172 OSlab]$ vi swap.sh
```

```
[CS24172 OSlab]$ sh swap.sh
```

```
Enter first number:
```

```
10
```

```
Enter second number: 20
```

```
Before swap first=10
```

```
second=20
```

After swap first=20
second=10

(iii) Greatest of three numbers:

Algorithm:

1. Start
2. Read three numbers from the user
3. Compare the first two numbers and then the third number
4. Find the greatest number
5. Print the result
6. Stop

Code:

```
echo "Enter the first number: " read a
echo "Enter the second number: " read b
echo "Enter the third number: " read c
if [ $a -gt $b ] && [ $a -gt $c ] then
echo "$a is the greatest"
elif [ $b -gt $a ] && [ $b -gt $c ] then
echo "$b is the greatest" else
echo "$c is the greatest" fi
```

Output:

```
[CS24172 OSlab]$ vi gíeatest.sh
```

```
[CS24172 OSlab]$ sh gíeatest.sh
```

```
Enter the first number:
```

```
10
```

```
Enter the second number:
```

```
30
```

```
Enter the third number:
```

```
20
```

```
30 is the greatest
```

(iv) Factorial of a number:

Algorithm:

1. Start
2. Read a number from the user.
3. Calculate the factorial using the loop concept by multiplying and reducing the number.
4. Print the result
5. Stop

Code:

```
echo "Enter the number whose factorial has to be computed: " read num
```

```
fact=1
```

```
while [ $num -gt 1 ] do
```

```
fact=$((fact*$num))
```

```
num=$((num-1)) done
```

```
echo "The factorial of $num is: $fact"
```

Output:

[CS24172 OSlab]\$ vi factoiial.sh [CS24172

Oslab]\$ sh factoiial.sh

Enter the number whose factorial has to be computed:

5

The factorial of 5 is: 120

(v) Fibonacci Series:

Algorithm:

1. Start
2. Read a number from the user.
3. Print 0 and 1 as starting numbers.
4. Add numbers and print result till the number is reduced.
5. Stop

Code:

```
echo "Enter the number: " read n
a=0 b=1
echo "The fibonacci series is: " echo "$a"
echo "$b"
while [ $n -ne 2 ] do
c=$((a+$b)) echo
"$c" a=$b
b=$c
n=$((n-1))
Done
```

Output:

```
[CS24172 OSlab]$ vi fibonacci.sh [CS24172  
OSlab]$ sh fibonacci.sh
```

Enter the number:

7

The fibonacci series is:

0

1

1

2

3

5

8

(vi) Generation of Prime numbers:

Algorithm:

1. Start
2. Read a number from the user.
3. Check whether the given number is divisible by only 1 and itself.
4. If true, print "It is a prime number".
5. Else, print "It is not a prime number".
6. Stop.

Code:

```
echo "Enter the range: " read n  
echo "The prime numbers are: " m=2  
while [ $m -le $n ] do  
i=2  
flag=0
```

```
while [ $i -le $((m/2)) ] do
if [ $((m%i)) -eq 0 ] then
flag=1
break fi
i=$((i+1))
done
if [ $flag -eq 0 ] then
echo "$m" fi
m=$((m+1))
done
```

Output:

[CS24172 OSlab]\$ vi Prime.sh

[CS24172 OSlab]\$ sh Prime.sh

Enter the range:

15

The prime numbers are: 2

3

5

7

11

13

(vii) Menu-Driven Calculator:

Algorithm:

1. Start
2. Read a numbers from the user.
3. Select the operation to be done using the numbers.
4. Do the specified operation
5. Print the result.
6. Stop.

Code:

```
sum=0
i="y"
echo "Enter the first number: " read a
echo "Enter the second number: " read b
echo "1. Addition"
echo "2. Subtraction"
echo "3. Multiplication"
echo "4. Division"
echo "Enter your choice: " read ch
case $ch in
1)sum=$((a+b)) echo
"Sum: " $sum
;;
2) sub=$((a-b))
echo "Subtraction: " $sub
;;
3) product=$((a*b)) echo
"Product: " $product
;;
```



```
4) division=$((($a/$b)) echo
"Quotient: " $division
;;
esac
```

Output:

[CS24172 OSlab]\$ vi mdcalulatoí.sh

[CS24172 OSlab]\$ sh mdcalulatoí.sh

Enter the first number:

8

Enter the second number:

9

1. Addition

2. Subtraction

3. Multiplication

4. Division

Enter your choice:

2

Subtraction: -1

(viii) Verification of Armstrong Number:

Algorithm:

1. Start
2. Read a number from the user.
3. Separate the digits and calculate the powers.
4. Sum the powers.
5. Print the result.
6. Stop.

Code:

```
echo "Enter a number: " read c
x=$c
sum=0
r=0
n=0
while [ $x -gt 0 ] do
r=$(( $x % 10)) n=$((
$r *$r *$r)) sum=$((
$sum+$n)) x=$(( $x/10))
done
if [ $sum -eq $c ] then
echo "$sum is an armstrong number" else
echo "$c is not armstrong number" fi
```

Output:

[CS24172 OSlab]\$ vi Aímstíong.sh

[CS24172 OSlab]\$ sh Aímstíong.sh

Enter a number:

153

153 is an armstrong number

RESULT:

The study of shell programming was conducted and the programs were successfully executed .